Problem J. Journey

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

Ritmo and *Fisica* are building a tuning device. A tuning device consists of a sequence a in which every integer in $1 \sim n$ occurs exactly twice, and a **regular bracket sequence** s containing n pairs of brackets. A tuning device works correctly if and only if it satisfies the following conditions:

- Create one vertex for each index.
- For every pair of matching brackets whose indices in s are (l_i, r_i), add an edge between these two vertices.
- For every value in a, let (x_i, y_i) be the indices of its two occurrences, and add an edge between these two vertices.
- The procedure above produces one cycle containing all $2n$ vertices and all $2n$ edges.

Ritmo has already installed a. Now you need to help *Fisica* find a corresponding sequence s. It can be proven that such a regular bracket sequence always exists.

Recall that a bracket sequence is **regular** if and only if it can be constructed using the following rules:

- `()` is a regular bracket sequence.
- If A is a regular bracket sequence, then `(` $+ A +$ `)` is a regular bracket sequence.
- If A, B are regular bracket sequences, then $A + B$ is a regular bracket sequence.

Input

This problem contains multiple test cases.

The first line contains a positive integer T ($1 \le T \le 10^4$), representing the number of test cases.

For each test case:

- The first line contains an integer n ($2 \le n \le 2 \times 10^5$).
- The second line contains $2n$ integers a_i ($1 \le a_i \le n$).

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5, and that every integer in $1 \sim n$ occurs exactly twice in a.

Output

For each test case, output a regular bracket sequence s of length $2n$ representing a valid answer.

Example

standard input	standard output
5 2 1 2 1 2 2 2 2 1 1 3 1 2 3 1 2 3 4 1 2 3 4 3 2 1 4 7 3 6 7 1 5 6 2 2 4 3 4 7 1 5	()() (()) ()()() ()()()() ()()()(()())()